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A learner's journey towards a chemical engineering degree

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Abstract

The overall goal of any engineering program is to maximize the capacity of its graduates to succeed academically and professionally. We describe how a path can be designed for learners to proceed towards this goal and describe the rationale used to create its foundational steps. It begins with a summer bridge program for incoming students before entering university as first-year undergraduates. Since the prior knowledge of these learners is not uniform, the bridge program is designed to provide opportunities for them to become better prepared academically for first-year engineering. These students thus transition to university-level learning more smoothly. In their first year, students work in groups to tackle socially relevant projects through an integrated 13-unit course that is designed based on integrated learning, collaboration, problem-solving, community engagement, and communication. Since teaching and learning using this approach is unusual and challenging, the curriculum must be carefully designed and implemented with adequate resources in place, particularly for cohorts of more than 1000 students in our case who work in small four or five-member teams. In upper years, learning in chemical engineering is enriched by conducting discovery-based workshops where students work on engineering problems requiring the application of new mathematical concepts. Finally, we describe a hybrid method for testing and assessment, where learners take tests individually, following which they are also provided with the option to retake these tests in groups to promote collaborative learning. Retaking tests in teams enhances the ability of learners to reflect and learn from mistakes and promotes peer mentoring.

KEYWORDS

bridge program, collaborative testing, discovery-based learning, integrated learning

1 | INTRODUCTION

A curriculum must be designed to build on learners' prior knowledge, helping them develop new competencies and augment those they already possess. The traditional pedagogies used to deliver engineering curricula use text, lectures, and labs that lack many elements used in the practice of engineering.[1] Thus, not all engineering graduates are fully prepared to be employable engineers. Whereas chemical engineering graduates must practice in increasingly diverse workplaces where communication and teamwork are required along with technical

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competence, conventional teacher-centred approaches and standalone courses have not universally succeeded in delivering the required outcomes and attributes.[2]

It is possible to develop interactive and experience-based learning environments, even for math-based material, to overcome this deficit, for instance through inquiry[3] or project-based learning,[4] where students are given opportunities to provide input into how their learning occurs.[1] These approaches enhance learners' independence and collaboration, building the competencies that employers ask for but find deficient in graduates,[5] including their understanding of ethics and social implications.[6]

Models of integrated curricula range from traditional discipline-based, objective-driven, instructor-controlled models to interest-based student exploration.[7] Our approach to engineering education is intermediate between these bookends, where instructors provide scaffolding for learners to succeed as they explore solutions for increasingly open-ended projects. The methods used for curriculum integration include themes, projects, problem-based learning, and enabling the role of the student as a researcher in a team.[8] This holistic approach is relevant to an increasingly complex world since it allows learners to form connections between different curriculum components and subjects and integrate their prior knowledge with new concepts throughout each stage of their learning.[9,10]

Because success is a multiplicative of ability and motivation,[11] an integrated problem- or project-based implementation that is transparently applicable to the challenges facing society renders a subject more attractive and accessible to a diversity of learners. This is also relevant from an equity, diversity, and inclusion perspective. For instance, there is evidence that men relate much more readily to traditional examples and analogies in engineering education than do women.[12] Hence, when new, integrated applications are related to real-life challenges, engineering topics become accessible to all students equally.

Learners' prior knowledge plays a critical role in their future learning.[13] Indeed, there is a substantial correlation between their interest in particular subjects and prior knowledge.[14] Hence, in our framework, prior knowledge is reinforced through a summer bridge program, where past learning is reviewed and linked through an introductory view into the first-year curriculum. Then, integrated learning in a 13-credit first-year curriculum common to all engineering students improves learners' appreciation of the relevance, context, and utility of their learning. Consequently, when they select chemical engineering as their degree program of choice (at the end of their first year as undergraduates), they can do so with greater understanding and confidence that their degree will allow them to address the needs of engineering and society throughout their career.

Collaborative learning and related assessments also improve student outcomes.[15] This form of learning leads to a convergence that produces an equivalence of knowledge among collaborators since learning is shared. The knowledge equivalence leads to learners becoming more similar to their learning partners regarding the extents of their knowledge, while their shared knowledge allows them all to gain the same understanding of concepts as their learning partners.[16]

Student engagement is also improved by collaborative learning. When they present collaborative problems, faculty provide students with active learning experiences, which induces student engagement in class discussions, projects, and beneficial interactions with their professors and peers.[17] These supportive relationships for students with mentors are important for their professional development and academic success.[18]

Here, we describe a structured framework (Figure 1) through which learners' prior knowledge is reinforced, which more readily enables their future learning and helps them develop and enhance competencies through an integrated and threaded curriculum. Since learning is inhibited by threats, such as high stakes testing,[19] and enhanced through challenges[20] and collaborative

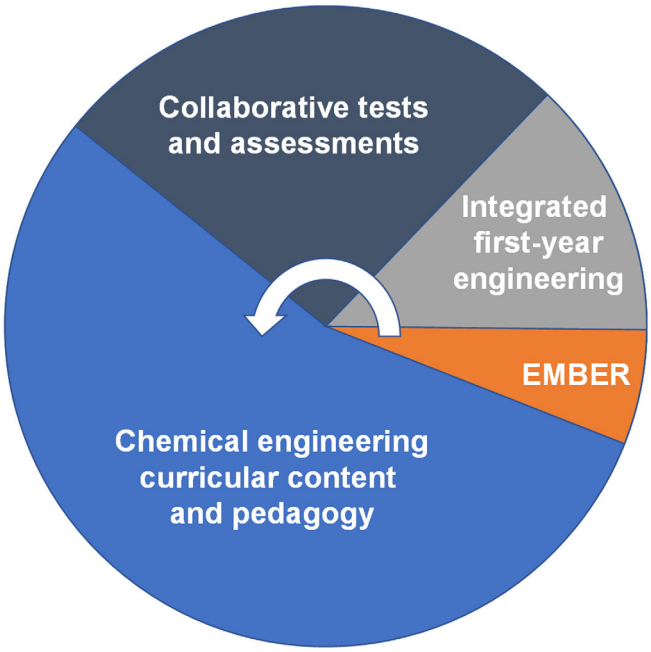


FIGURE 1 Schematic of the roadmap with the various steps taken with the goal of improving learning. The process starts before students join the University with a summer bridge onboarding program (EMBER, Engineering Mentorship & Bridging Education Resources)

learning,[21] the curricular framework poses challenges through team projects that become progressively more open-ended.

2 | BEFORE ENTERING UNIVERSITY: SUMMER BRIDGE PROGRAMS

2.1 | The idea: Rationale and structure

The transition from high school to university can be challenging for students.[22,23] In response, an increasing number of universities and colleges offer summer bridge programs to help students entering post-secondary education.[24,25] Summer bridge programs can take various forms ranging from eight weeks to a few days.[25] While some bridge programs are geared towards remediation of students who performed poorly in high school, other programs target students who require a review of prerequisite academic content.[26,27] Despite the varied nature of summer bridge programs, most programs offer both academic and social resources designed to support student academic achievement as well as their mental health.[25,26,28,29]

Several studies have shown a positive correlation between academic success and participation in a summer bridge program. Black students who enrol in a summer bridge program had a significant increase in GPA and retention rate compared to students who did not participate in the program.[30] Moreover, students who participated in the program were 10% more likely to graduate from university.[30] Similarly, another study found that students who participated in a summer bridge program had a significantly higher retention rate and outperformed their peers in their first-year history, math, and English courses.[26] Summer bridge programs have been known to increase student retention into the second and third years and improve the graduation rate after four years.[29]

Beyond academic success, summer bridge programs also help students feel more prepared for post-secondary education.[25] Several studies have shown that students' sense of preparedness significantly increased after participation in a science, technology, engineering, and mathematics (STEM) summer bridge program.[31–33] Interestingly, STEM students who participated in summer bridge programs also reported a greater sense of belonging[33] and are more likely to graduate from a STEM program and pursue a career in STEM.[34]

Not all studies showed a prolonged positive effect of summer bridge programs. A study at eight post-secondary institutions in Texas found that summer bridge programs only improved student's academic success in the first year and had no significant long-term impact on academic performance and retention after the first year.[28] Similarly, a study at St. Mary's University showed no significant academic benefits for students who participated in the summer bridge program.[35] Nevertheless, evidence supports that summer bridge programs can be beneficial to students' academic success, increase student retention rates, and improve students' sense of belonging to the community as well as their discipline.

Of note, several studies have shown that summer bridge programs significantly improve the academic success and retention of underrepresented students, defined by demographics based on race or socioeconomic status.[26–30] One showed that participation in a summer bridge program improved the retention of underrepresented students in a STEM discipline by up to 10%.[33]

Given these benefits, summer bridge programs have taken on greater relevance during the COVID-19 pandemic, since many students have felt disconnected from their cohorts and underprepared for post-secondary education. In light of these challenges faced by students, McMaster Engineering developed a summer bridge program called Engineering Mentorship & Bridging Education Resources (EMBER). It is designed to support the transition of all incoming engineering students into their first year of studies by preparing them academically and improving their sense of belonging.

EMBER offers three asynchronous online courses in math, physics, and chemistry made available to students to study at their own pace. Each module, approximately 45 min long, includes videos, virtual simulations, tests, formula sheets, and other helpful resources. The courses cover the prerequisite high school content required for the first-year engineering curriculum. The course content was developed by nine learning resource assistants (LRAs) and finalized after review by a faculty member. The LRAs are undergraduate engineering students from various departments, ranging from students who recently completed their first year of studies to recent graduates. The LRAs developed over 140 videos containing over 16 h of educational content that includes 20 modules that cover 75 topics spanning the three subjects. These modules and materials are available to first-year students during the entire academic year.

In addition to the asynchronous online courses, EMBER offers daily synchronous online tutorials for two weeks in August. The tutorials are led by the LRAs who engage with students by answering questions. As well, an online community was established by encouraging participants to interact with one another virtually over MS Teams. The tutorials reinforce the concepts covered in the online modules and introduce extensions to those topics. Each tutorial is 55 min in duration and is offered

nine times every day for each subject to accommodate students' schedules, particularly for international and out-of-province students who live in different time zones across the world. The tutorials are recorded so that students can watch them asynchronously in case they have missed the live offering or desire to revisit the content later.

2.2 | Results and student satisfaction

2.2.1 | Student engagement statistics

The engagement of the incoming first-year students who participated in EMBER was overwhelmingly positive. Overall, 942 unique students registered for EMBER, representing over 86% of the incoming class. Of those who registered for the program, about 600 students accessed and used the online modules through the McMaster University Learning Management System (Avenue-to-Learn). Over 550 students attended at least one live tutorial. Approximately 250 students actively engaged in the live tutorials throughout the two weeks.

The Engineering Outreach team led a campaign to engage incoming students through videos, social media, and e-campaigns. The EMBER Ambassador Program was developed to provide opportunities for members of the incoming class to participate in peer-to-peer outreach. Four students were selected from 25 applicants. These EMBER Ambassadors led takeovers of the McMaster Engineering Instagram account to share their experiences and engage their peers, which resulted in an average reach of 1900 individuals per post.

2.2.2 | Student feedback on academic preparation

After the program concluded, students were requested to respond to an anonymous survey and provide feedback. The overwhelming response was that EMBER helped them prepare for university and alleviate anxiety about the upcoming academic year (see Table 1). Of the 125 responses collected, 96% of respondents felt better prepared for their first year because of EMBER and a similar 96% would recommend EMBER to future first-year students.

2.2.3 | Student feedback on sense of belonging

EMBER was a platform for students to not only gain academic knowledge but also to facilitate connections with upper-year students. The LRAs played a crucial role in enhancing the student experience. From the survey, 94% of the students found the LRAs helpful and many students emphasized their appreciation of how the LRAs made them feel welcomed.

Overall, the feedback shows EMBER was valuable for supporting students during the COVID-19 pandemic. It is a resource by which incoming students can review their prior learning and the program alleviates anxiety related to their transition to university. Second, EMBER provides a link for incoming students to engage with upper-year students, allowing them to establish a network of students, staff, and faculty who they can later rely on to help bridge their transition from high school to university.

There are some aspects of EMBER that can be further explored through longitudinal studies over multiple years and in multiple cohorts. To gauge the impact of EMBER on academic success, additional data must be gathered about the academic performance of students who enrolled in the program and compare it with that of students who did not participate. The program's impact on student retention can also be measured by comparing the retention rates of students who participated in EMBER with those who did not.

TABLE 1 EMBER (Engineering Mentorship & Bridging Education Resources) student survey responses

Question	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
I feel EMBER has made me feel more prepared for first-year courses.	1	0	4	56	64
Online modules were engaging and explained concepts well.	1	1	3	61	59
LRAs were helpful and explained concepts well.	0	2	5	47	70
I would recommend EMBER to someone who is entering the first year.	0	3	1	25	94

3 | FIRST YEAR ENGINEERING: INTEGRATED LEARNING AT A LARGE SCALE

3.1 | The idea: Learning from experience

Practicing engineers are routinely asked to apply their content knowledge and professional skill set to provide solutions for diverse problems. However, mapping the technical and personal abilities required of engineers by industry to curricula have revealed significant shortcomings in their education.[36] Typical engineering curricula are tightly focused on engineering science and technical content, a process called academic drift,[37] resulting in insufficient integration of topics or their relation to industrial practice. Programs are content-driven and consequently do not provide sufficient design experiences for students or bolster key communication skills and teamwork. It is now broadly recognized that engineering programs must develop greater awareness among students of the social, environmental, economic, and legal issues that are part of the reality of modern engineering practice.[38]

Since real-world problems (particularly in current global multidisciplinary industrial environments) do not occur in neat disciplinary categories, engineers must be able to synthesize knowledge across disciplines. In addition to their field-specific technical knowledge, they should understand and apply concepts from several disciplines to create solutions for complex problems. Overall, today's complex, integrated systems require engineering graduates with a more diverse skillset and a greater ability to communicate with their colleagues from various disciplines than their counterparts two or three decades ago.[39,40]

To prepare graduates for these new needs, many postsecondary institutions now offer students opportunities to practice authentic problem solving through community-engaged design initiatives that support engineering students as they build their capacity for critical thinking, empathetic understanding, and ethical considerations for human problems.[41] Through participation in community-engaged design projects, students develop their capacity to generate ideas, think creatively, and innovate as they navigate resistance to change.[42] These hands-on learning experiences allow learners to develop the necessary skills to solve problems ethically and sustainably as a way to contribute to the health of society.

Engineering degree programs are therefore integrating their focus on fundamentals with rigorous, academically appropriate hands-on experiences with real-world applications. Over two decades ago, the University of Massachusetts Dartmouth (UMD) piloted an integrated, first-year engineering curriculum that taught 31 credits across two semesters. The program was modelled after successful undergraduate experiments at other universities, such as those in the National Science Foundation (NSF) Coalition and Rensselaer Polytechnic Institute. The new UMD program integrated introductory sequences in physics, calculus, chemistry, English, and engineering. Teaching and teamwork among students and faculty occurred in a specially designed technology-enabled classroom using active and cooperative learning methods.

Faculty interested in implementing an integrated curriculum must first answer the question of why an integrated curriculum might offer an improved learning experience for entering engineering students. Constructivist learning theory suggests that learning occurs as students add elements to their ideological scaffolding through experiences.[43] Students construct, discover, transform, and extend their knowledge by interacting with their environment and extracting meaning from their experiences. Learning is something the learner does, not something that is done to a learner. In general, students do not passively accept knowledge from the teacher or curriculum. Rather, they use new information gained from experience to activate their existing cognitive structures or construct new ones.[44,45]

The role of the educator is to create conditions within which students can construct meaning from new material, study by processing the material through existing cognitive structures, and then retain it in long-term memory where it remains open to further processing and possible reconstruction. It is crucial, in this sense, that the experiences students have are, in fact, educative. Educators are tasked with providing students with quality experiences that will be applied to later experiences.[46] All of these factors should be considered for any integrated curriculum.

In addition to the important consideration of the quality of the experiences provided to students, the first-year curriculum has two essential purposes:

1. To create a strong foundation for upper-year courses. To motivate young learners early on, they must learn to better understand what engineers do, how they do it, and why engineering work matters. Hence, first-year students should be exposed to different engineering disciplines to better explore their interests and curiosity so that they can wisely choose the discipline that fits their goals and aspirations. They also require the technical foundation and communication skills to be successful in upper-year courses.[47]
2. To help students through the transition from high school to university. This change can be unsettling for young learners since the transition from high school to university can place significant demands on them.

University life can be demanding and stressful for a new student and requires higher levels of independence, initiative, and self-regulation.[22,23]

To address these needs, McMaster University has altered its first-year engineering curriculum by developing a new 13-credit course, ENGINEER 1P13, 'Integrated Cornerstone Design Projects in Engineering', taught through both Fall and Winter semesters. ENG 1P13 is an amalgamation of four previously existing first-year courses that covered computation, materials, graphics, design, practice, and the profession. The new course is part of a larger curriculum transformation at McMaster called The Pivot to prepare students to be flexible in a rapidly changing world and meet challenges not yet imagined.

3.2 | Structure of the experience

ENG 1P13 is a large course offered to more than 1100 incoming first-year students for the first time in Fall 2020. It involves eight instructors from different engineering disciplines, 10 instructional assistant interns (IAIs), more than 150 teaching assistants (TAs), and lab technicians and staff.

This integrated full-year course employs a project-based experiential framework for design thinking that focuses on foundational engineering knowledge through five project modules. Consistent with Dewey's focus on continuity of learning through experience, each project transitions from well-structured to open-ended and each new module requires students to apply prior learning gained through previous modules.[46]

Each episodic experience is followed by a reflection on what the experience meant and how it might be applied during the next module. Students gain insights and empathy for those whose problems they are working to solve. Technical skills and subject expertise steadily build upon one another for engineering design and graphics, programming, professional engineering practice, and the structure and properties of materials. The final project of the course is entirely open-ended and reinforces all prior lessons learned and is designed to facilitate increasingly complex problem solving through design thinking, client-focused design approaches, and social consciousness.[48]

Students have 11 h of contact time with the instructional team per week through 3 lecture hours, 2 h of design studios, and 6 h of labs. Lectures, delivered by faculty members, introduce the theory and concepts to students. Design studios are led by an IAI and supported by TAs and lab technicians. At least one faculty member in each design studio plays the role of mentor for students.

Student work in design studios is designated as project time when student teams work collaboratively towards milestones for a project. In addition to lectures and student work in design studios, there are two 3-h labs each week led by an IAI who is supported by other IAIs, TAs, and lab technicians. A computer lab provides hands-on experiences through which they learn computing and graphic design while another lab focuses on materials science experiments. In some labs, students complete individual activities related to project milestones.

Through this structure, every first-year student develops a relationship with at least one faculty member, their upper-level peer mentors, and their peers.[48] George Kuh would consider this course to consist of high impact practices (HIPS) that have been shown to increase student engagement and success, particularly among students who face barriers to succeed in post-secondary education. Providing opportunities for students in this manner to discuss their coursework with faculty members, their support team, and peers enables frequent feedback to them regarding their learning. As students complete their learning tasks while being mentored by the teaching team, they are able to connect the new curricular material with their prior knowledge and also develop the necessary professional insights and skills required of engineering graduates.

Time spent in design studios allows students to work on different project milestones with their team members. The main focus of labs is to teach computing and computer-aided design (CAD) skills, as well as introduce students to engineering science and communicating content through hands-on experiences. The lab and design studio spaces are oriented to answer the fundamental questions about the nature of engineering, that is, what engineers do, how they do it, and why it matters.

Understanding their suppositions and testing them through engaging hands-on and academically rigorous projects provides a strong motivation for learners and helps them connect theory to real-life challenges and applications. The integrated curriculum augments the relevance between the material being learned and students' perceptions of their career needs as future engineers. An important goal of the integrated curriculum is to motivate students to develop social consciousness by mastering the material and applying it for the benefit of society.

When courses are integrated, a coordinated teaching team can avoid haphazard repetition of material and concentrate redundancy in conceptual areas where students have demonstrated difficulty in mastering the material.[49] This allows the teaching team to practice and reinforce difficult topics in the single integrated course rather than leaving it to introductory levels in several

courses. This approach results in better retention of the material, which smooths the transition between subjects.

3.2.1 | Overview of projects/modules

The 13-credit 1P13 course consists of five sequential student projects, of which each of the latter four builds on student learning in the previous project. Project topics range from autonomous vehicles, to renewable technologies, to healthcare, to sustainability and community, all tied to real-life challenges relatable to aspiring engineers. The course enables continuous learning for computing, graphic design, material engineering, and engineering practice and profession.

In each project, students learn new concepts in these topics and combine them with their prior learning. By doing so, students become prepared to complete the fifth and final open-ended client-based project, where they define their problem statement and select their tools for project design and solution.

The five projects enhance learners' technical competence and creativity, help them learn how to work with diverse problem definitions and possible solutions, and have them consider the relevant business, ethical, cultural, and social considerations. Studies show there is lower student attrition when the curriculum links concepts to real-world practice, where active and collaborative learning improves learners' performance and their retention of information over traditional methods.[50]

3.2.2 | Teamwork structure

In ENG 1P13, a group of four or five students is randomly assigned to a team for each project following a block-schedule format, that is, all students in that team attend the same lectures, design studios, and labs. Learners' gender and their anticipated administrative and technical roles in the team are also considered in group formation.

For each project, students must prepare a team charter where they formally document their team's personnel and the administrative roles and responsibilities each member will assume for the duration of the project. At the beginning of each project, the students agree as a team about who will take the lead for each administrative task: manager, administrator, coordinator, and subject matter expert. Students are required to change their roles for each successive project so by the end of the course they have all experienced the responsibilities for each of the administrative tasks. At the end of each project, students evaluate their group members by providing feedback through a peer-evaluation tool.

3.2.3 | Workshop series: Technical communication and reflection

In addition to learning technical content, students participate in workshops to learn fundamental skills to conduct research, write a report, present their work, and communicate their thoughts to others. Students attend four 3-h design communication workshops that educate and train them in three different themes: research resources, written communication, and oral communication. There are another two 3-h workshops that introduce students to the process of reflection through activities that train them to learn from their experiences in the course, a useful opportunity for them to analyze their learning process and explore how they learn.

3.2.4 | Learning portfolio

Each team member must set-up an online learning portfolio to document their project work. To create and populate their online learning portfolio, students use a cloud-based web development platform that allows users to create a website through online drag and drop tools. Students document their project journey and record what they do and what they learn throughout the course. The portfolios are only graded in Pass/Fail format upon completion. However, students are encouraged to record as much detail as they are able since this documentation becomes part of the portfolio that they use during their final reflection workshop to remember and reflect upon specific aspects of their course experience.

The role of reflection is fundamental in all experiential education courses to generate learning by connecting theory to practice, deepen it by challenging learners' thinking, and document it by producing tangible expressions of the new understanding they gain by integrating experience and disciplinary knowledge.[51] Woven within this course design, learners' reflections are meant to support them as they make meaning out of their learning from each module that is then applied to the next learning element. Insights and deeper understanding of foundational content thus build upon one another rather than having learners simply go from one task to the next without further examination and reflection.

3.2.5 | Communication and obtaining feedback from people outside engineering

A part of one of the four projects in the course is offered in partnership with the IMPACT (Interdisciplinary, Mentorship, Practice, Community, Transformative) initiative

at the university. IMPACT engages undergraduate and graduate students from various faculties, alumni, healthcare partners, and real clients to understand and address challenges experiences by people living with disabilities. ENG 1P13 students present their design ideas to this group and can obtain feedback from others who have different points of view.

In the project, students learn to integrate their conceptual and applied knowledge of science and engineering to collaboratively create a design that helps a real client. The project milestones require the engineering students to create prototypes of their design and present their work to the IMPACT group through two scheduled design reviews. This interdisciplinary exchange of ideas creates meaningful experiences for better design, which results in positive community engagement and creates pathways for change.

4 | IN-CLASS WORKSHOP MODULES AND DISCOVERY-BASED LEARNING

Research has shown that active learning strategies, such as those that incorporate lecture activities and breaks from the traditional 'sage on the stage' approach, have positive impacts on student learning and success.[52] Moreover, the combination of in-class activities and out-of-class readings through the use of notes-with-gaps and other methods has been shown to improve engagement and retention of class material.[53–55]

4.1 | The idea

Recent observations in our departments have shown that courses rich in mathematical concepts to solve applied problems (a.k.a. the math courses, not to be confused with engineering courses) are often regarded by the students as dry or boring, typically leading to issues with class attendance, engagement, and student success. For example, numerical methods problems focused on solving linear algebraic equations often present coefficient matrices or systems of equations, where variables (x) are presented without context. These systems are important for learning the core concepts of linear equation systems (independence, degrees-of-freedom, etc.) but often do not bridge the gap between theory and application.

We contend that establishing context through real engineering examples is a much more effective way to engage students. In their traditional form, many courses suffer from having high activation energy. Students resist the material (e.g., solving large systems of linear or nonlinear equations) unless they are provided real, meaningful learning experiences and can see the power of what they are learning instead of simply being bombarded with arcane mathematical concepts (e.g., solving the mass balances of a large process flow sheet, or using a computer algorithm to perform a fugacity balance for a non-ideal mixture).

As a specific example, consider the linear equation system presented previously. If the problem is established as a steady-state discretized energy balance, such as one would require for finite element analysis, the problem of solving a large system of equations for a set of numbers quickly becomes one of solving for the temperature distribution within a control volume, which opens up an opportunity for discussion around engineering design. This is the strategy we take to overcome the activation energy barrier in our program.

We use a series of in-class workshop modules to turn the classroom into an open discussion of engineering problems that benefit from the application of mathematical concepts, thus focusing on the outcome of learning the method as a part of learning the concept itself. These problems are solved using a variety of software programs such as MATLAB (to also teach the core concepts of applied programming to undergraduate students) and more industry-relevant packages like Microsoft Excel and GAMS (used extensively as a set-up language to solve industrially relevant optimization problems).

So far, we have developed course workshop modules for two courses in the curriculum: Numerical Methods and Computing for Chemical Engineers (taught at the second level) and Applications of Optimization in Chemical Engineering (taught at the fourth level). The modules were developed with the help of numerous students that have experienced these courses and contributed to the workshop modules alongside the instructor to provide the student perspective, which allows for tricky or abstract content to receive the attention it deserves and to put an appropriate emphasis on the applications and uses of concepts (student perspective) while still maintaining the rigour and completeness desire by the instructor.

The workshops include a blend of content (written much like an abridged textbook focused on knowledge transfer rather than knowledge demonstration), notes-with-gaps, and active learning activities and problems woven together into one comprehensive custom courseware. The modules are released for free as PDFs to the students. These workshop modules are constantly improved and expanded upon and act as a comprehensive resource for the students. Potential benefits of this classroom approach include (but are not limited to) the following:

- Accessibility: These modules provide students with a free resource for notetaking and practice that covers all the content in the course. This is of benefit for students with financial limitations because they do not need to buy a textbook. Moreover, due to the open-book examinations used in these courses and many others in the department, the modules provide a go-to reference for class material and permit the use of assessments that expand on core concepts rather than simply having the students apply them from memory.
- Exploration-based learning: These modules give the instructor the chance to spend large portions (roughly 50%) of class time taking up ideas, chatting with students one-on-one, and facilitating discussions. These activities are gratifying for the students and give them a sense of accomplishment and identity with the material. Moreover, these activities can lead to discussions on topics not explicitly covered in the modules themselves, also giving students ownership of their learning by expanding discussions to topics they find interesting.
- Clear expectations: One of the greatest challenges for students is that they expect plug and chug problems. Instead, reaching the higher tiers of Bloom's taxonomy often requires open-ended projects or (in an examination scenario) that requires the students to explain or modify an existing concept to a new scenario they have never seen before. The modules greatly assist the instructor in communicating this expectation because they allow practice on these open-ended problems in class as a team and with no risk of failure.

4.2 | Results and student satisfaction

To quantify the effectiveness of these modules, feedback from students on their opinions of the modules as learning tools has been collected and tabulated. Aside from the written feedback from students that is used to continuously improve the modules, data is available outlining the perceived effectiveness of the modules in and beyond the classroom. The data is collected by requesting the students to rank their experiences with the modules on a Likert scale for a variety of questions.

Shown in Figure 2 are the results for students rating the helpfulness of the in-class workshops for their learning experience in a fourth-year Chemical Engineering course on Optimization (ChE 4G03).[56] Figure 3 shows the data for the same class rating their preference for using the modules over board notes or lecture slides.

Figure 2 shows that students strongly value the structure of the in-class workshops and modules, especially during lecture times. Moreover, Figure 3 indicates that

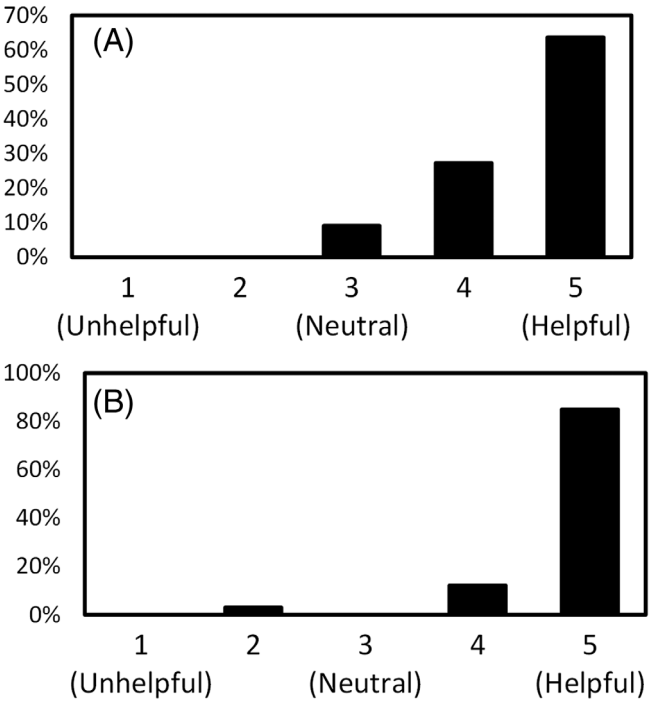


FIGURE 2 Student ratings of the usefulness of the ChE 4G03 (optimization course) workshop modules for exam preparation (panel A) and during lecture (panel B). Data based on two cohorts of approximately 70 students each. This data is an extension of previously presented results[56]

the majority of students strongly feel that these workshop modules are superior learning tools and promote a better learning environment than lectures that use presentation slides or black/whiteboard notes. This data motivated the development of the modules now used in Numerical Methods and Computing (2E04), the results of which are presented in Figures 4 and 5.

The modules in numerical methods and computing are also regarded highly, with over 80% of students indicating that the modules were 'very useful' or 'useful' both for exam preparation and for notetaking and engagement in lectures. Moreover, over 80% of students agree that the modules are superior to lecture slides or board notes as a vehicle to promote their learning in class.

5 | COLLABORATIVE TESTING AND ASSESSMENT

5.1 | The idea

Since teamwork and collaboration are key competencies of the engineering profession, most undergraduate programs strive to incorporate these into mainstream

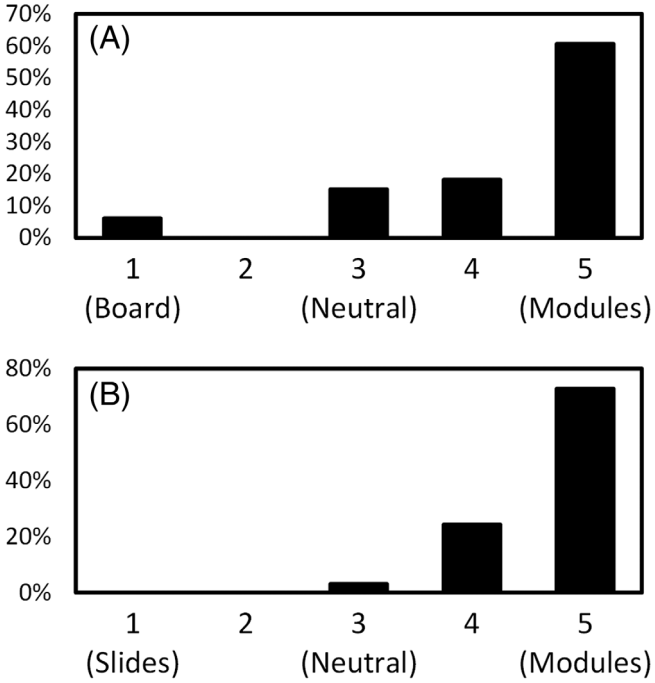


FIGURE 3 Student preferences for the ChE 4G03 (optimization course) workshop modules relative to board notes (panel A) and presentation slides (panel B). Data based on two cohorts of approximately 70 students each. This data is an extension of previously presented results[56]

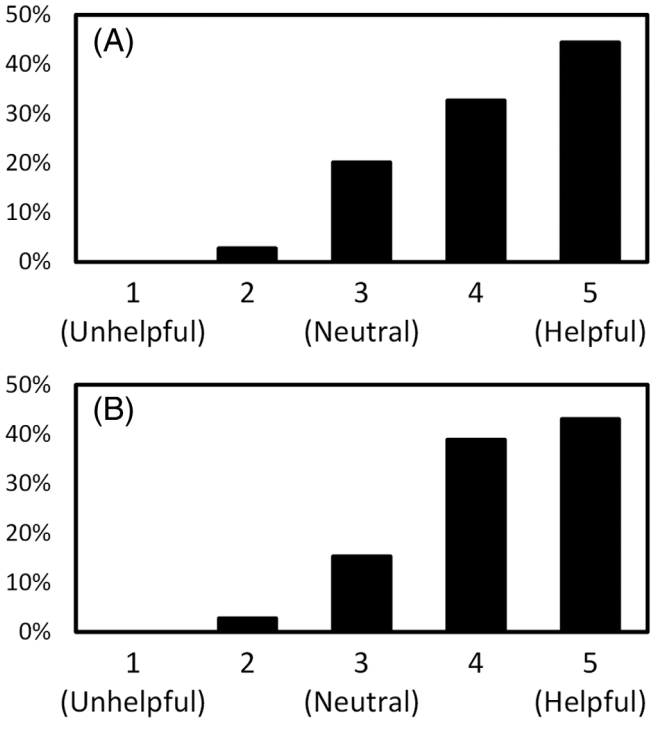


FIGURE 4 Student ratings of the usefulness of the ChE 2E04 (course on numerical methods) workshop modules for exam preparation (panel A) and during lecture (panel B). Data based on one cohort of approximately 160 students

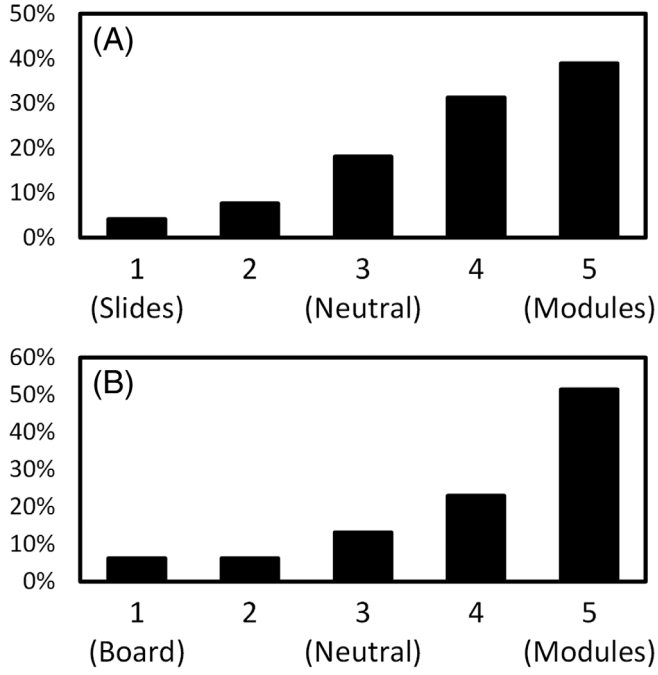


FIGURE 5 Student preferences for the ChE 2E04 (numerical methods) workshop modules relative to board notes (panel A) and presentation slides (panel B). Data based on one cohort of approximately 160 students

assessments of student performance, such as tests and exams.[57–59] Tests or exams often elicit anxiety in students, which interferes with their learning due to factors within (e.g., time management, preparation) and outside (poorly written or culturally biased questions) their control.[60,61]

Collaborative assessments have been shown to provide some relief from this anxiety.[62,63] However, it may still be important to extract each student's individual understanding of course material through a given assessment. Project-based courses that only include collaborative deliverables can sometimes fail to provide students with the opportunity to demonstrate their grasp of key concepts. Collaborative re-testing is a method that produces a record of student performance on an individual level while also offering students a low-stakes way to improve their grades by learning from each other and discussing problems in an active learning setting.[64–66]

Collaborative re-testing follows these general steps:

1. Students are given a midterm, quiz, or other assessment in the typical fashion. These assessments are completed individually under rules that the instructor has deemed appropriate.
2. Students are given one to two days to reflect on their assessment and discuss the problems with their classmates.

3. During the following lecture, students are allowed to join, but not obligated to participate in, randomly assigned teams of four to five classmates. Each group is given the same assessment to complete collaboratively for the duration of the lecture (typically an hour).
4. During the grading phase, a portion of the exam re-written by the team counts towards the true score of a student for that assessment. Thus, if the group re-write has a higher evaluation than the original individual answer, the student gains a small number of points towards their final grade. If the group re-write is worse than the original individual assessment, it is omitted from the grade calculation. Typical ratios of individual-to-group weighting schemes are 85%:15%.

5.2 | Benefits of collaborative testing

As will be made clear in the results and discussion to follow, students and instructors appreciate the concept of collaborative testing for several reasons:

- Redemption: Students who struggle on an individual test are offered a chance to redeem themselves and improve their grades by recovering points via the collaborative re-test. By the nature of the grading scheme, the worse a student performs on the individual test, the more they stand to gain by participating—and succeeding—in the group re-write. This is a critical incentive because it often pushes a student to study and prepare for the areas in which they struggle. After all, they know exactly how they can recover points on their test or midterm if they prepare correctly. This incentive of redemption is often high enough to engage struggling students in group discussions and, by extension, fosters additional engagement with the material.
- Learning from mistakes: Students often feel high levels of anxiety before, during, and immediately after a test or exam. These feelings are often brought on by the threat of not achieving a high grade. Many instructors would prefer that the students focus on learning rather than assessment, but this connection is often lost because students (or if we are being honest, everyone) place a huge emphasis on grades as a measure of self-worth. By participating in the collaborative re-write, students have the chance to face mistakes, learn from them, and positively build on them. Instead of seeing what they did wrong and regretting that they have lost those points forever, they see what they did wrong and can discuss why with teammates. Then, they have the chance to recover points via redemption, leading to a positive feeling and experience associated with making a mistake: the students learn from it, demonstrate that learning, earn points for that growth, and can move on with a better attitude.
- Mentorship: Surprisingly, many students who perform very well on the individual writing of the assessment will participate in the collaborative re-write anyway. Although these students do not often gain many points on the assessment by doing so, it provides a sense of confidence that they did indeed get the correct answer the first time around. Moreover, when these top-achieving students are paired with struggling students, the achievers have an opportunity to mentor the struggling students, benefiting both in the process: the struggling student learns more about the material and stands to gain points on their final grade (as mentioned), and the high-achieving student feels a sense of pride having assisted another student while reinforcing their understanding of the material at the same time. Indeed, it has been shown that students are fully capable of explaining abstract concepts to their peers.[63,67]
- Accomplishment: Not to be confused with redemption, there is something to be said for walking out of a test feeling like you nailed it. A feeling of satisfaction and success can lead to improved motivation and a positive attitude regarding course content. By allowing students to re-write an assessment in teams, the feeling of satisfaction having likely achieved a high grade typically results in positivity and a sense of accomplishment for students.
- Concept reviews: Instructors can use the in-class rewrite of an assessment as a tool to review the assessment instead of the usual act of taking up the questions in class or during office hours. The collaborative rewrite reaches nearly every student in the class and allows the students to review concepts with each other, thereby improving the efficiency of these discussions and preventing the instructor from revisiting old concepts since the students are motivated to review them on their own before and after the collaborate rewrite.

5.3 | Results and student satisfaction

What follows in Figure 6 are the results from 1188 midterm scores from a variety of second-, third-, and fourth-year courses in the Department of Chemical Engineering at McMaster since 2017. The scores reported in Figure 6 are for assessments in which a collaborative re-write was offered within two days of the original assessment. Of the 1188 individual scores, 984 scores were recorded for the collaborative rewrite of the tests.

Several observations can be drawn from this data. First and foremost, the scores for the collaborative rewrite of all tests (aggregated across all courses) are

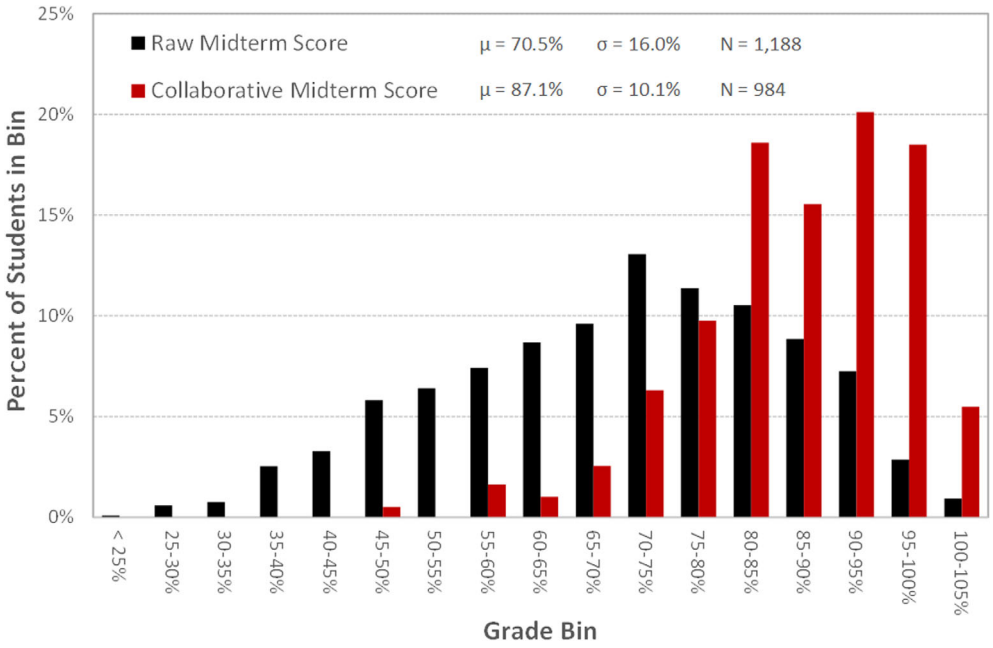


FIGURE 6 Individual midterm scores (black) and collaborative rewrite scores (red) for several courses in chemical engineering since 2017

generally much higher and demonstrate a tighter distribution centred around 90%. The individual scores are much lower (average of 70% vs. 87% for the rewrite scores) and considerably more diverse (standard deviation of 16% vs. only 10% for the rewrites). Of course, the higher averages have a great deal to do with the fact that the students have seen the problems and are tackling them in teams, but the results are unmistakable: the average student performs 17% better, with 770/984 (78%) of students attempting the rewrite getting over 80%. Over 5% of students get even greater than 100% on the collaborative rewrite due to the typical presence of a bonus problem.

Participation is very high (984/1188 or 83% of eligible students chose to participate in the rewrite) even though more than 30% of students achieve over 80% on the individual score anyway. These students stand to gain on their overall test performance from the rewrite, but most participate anyway, leading to the conclusion that they appreciate the chance to mentor struggling students and feel a sense of accomplishment for performing well on the assessment.

One may be concerned that the concept of collaborative rewrites of tests will lead to inflated grades. To show that is not the case, consider the data presented in Figure 6. Midterms in these courses are worth an average of 20% of the final course grade, with the typical course consisting of two midterms (totalling 40%) along with several assignments and a final examination, none of which include a collaborative re-write. Based on the data available, the average student can expect to see a roughly 2.5% increase to their test score by participating in the rewrite. When weighted against their final grade, this results in only a 0.5%–1.0% increase for the typical student if the course includes two midterms, which is not significant enough to lead to major grade inflation issues. Moreover, the students that struggle individually stand to gain the most, and thus the concept of collaborative rewrites tends to bring those students up (raise the floor, so to speak) while high achievers tend to only see very modest increases to their grade if any.

6 | CONCLUDING REMARKS

The engineering curriculum must evolve continuously to address the needs of society, employers, and technological advancements. We make a case for conducting responsible and robust experiments based on ideas from pedagogical theory, psychology, and cognitive sciences that reveal how people learn.

While our curricular design seeks inspiration from past scholarship, its implementation has occurred over a unique historical period. COVID-19 has created enormous challenges for society and higher education institutions. It has become even more critical to ensure a smooth and supportive transition of learners from high school to university. We address this challenge through a new McMaster summer bridge program, EMBER. While we report initial student engagement and satisfaction, longer-term monitoring of learner outcomes, such as their success, retention, and employability, is in place to more fully determine the impact of this program.

Engineers must become more adept at integrating knowledge from multiple disciplines to develop creative solutions for socially important problems. The best place to begin this learning is during the first year of an engineering degree. However, the first undergraduate year is also the most challenging for engineering curricula due to its necessary versatility and wide backgrounds of incoming students. Students have limited knowledge of the variety of engineering fields and applications and they initially require well-structured problems to work on.

The undifferentiated overall first-year engineering student cohort is also typically very large. We describe how a 13-unit course created to overcome these challenges was offered for the first time in Fall 2020. Several follow up long-term research projects are in place to obtain formative and summative evaluations of the outcomes of this new experience for learners. While the literature supports the use of this method of integrated teaching and learning, the size of our experiment with a cohort of more than 1000 engineering students is unprecedented.

There is also plenty of room for innovation in the upper years of chemical engineering programs. We provide two examples implemented at McMaster. The first centres on discovery-based learning, where students learn mathematical concepts by applying them to processes used in chemical engineering practice. This provides a strong context for learners to understand why math is required and the use of this approach leads to learning at the higher tiers of Bloom's taxonomy.

The second example rethinks testing and assessment by allowing learners the option to include a team-based retake of a test after they have completed it individually. This approach has many advantages, such as providing learners with the opportunity to reflect on their work. The ability to reflect and learn from mistakes is invaluable for an engineer. Learners, including engineering students, can continuously improve themselves through reflection. Hence, designing and implementing opportunities for our students to reflect, making this an essential part of their education, should serve the engineering profession well in future.

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